

CHAPTER 2: ASSET CLASSES AND FINANCIAL INSTRUMENTS

2. Money market securities are called *cash equivalents* because of their high level of liquidity. The prices of money market securities are very stable, and they can be converted to cash (i.e., sold) on very short notice and with very low transaction costs. Examples of money market securities include Treasury bills, commercial paper, and banker's acceptances, each of which is highly marketable and traded in the secondary market.
11. a. At $t = 0$, the value of the index is: $(90 + 50 + 100)/3 = 80$
At $t = 1$, the value of the index is: $(95 + 45 + 110)/3 = 83.333$
The rate of return is: $(83.333/80) - 1 = 4.17\%$
- b. In the absence of a split, Stock C would sell for 110, so the value of the index would be: $(95+45+110)/3 = 250/3 = 83.333$ with a divisor of 3.
After the split, stock C sells for 55. Therefore, we need to find the divisor (d) such that: $83.333 = (95 + 45 + 55)/d \Rightarrow d = 2.340$. The divisor fell, which is always the case after one of the firms in an index splits its shares.
- c. The return is zero. The index remains unchanged because the return for each stock separately equals zero.
12. a. Total market value at $t = 0$ is: $(\$9,000 + \$10,000 + \$20,000) = \$39,000$
Total market value at $t = 1$ is: $(\$9,500 + \$9,000 + \$22,000) = \$40,500$
Rate of return = $(\$40,500/\$39,000) - 1 = 3.85\%$
- b. The return on each stock is as follows:
 $r_A = (95/90) - 1 = 0.0556$
 $r_B = (45/50) - 1 = -0.10$
 $r_C = (110/100) - 1 = 0.10$
The equally weighted average is:
 $[0.0556 + (-0.10) + 0.10]/3 = 0.0185 = 1.85\%$